



7957 - MEGA Spectra: Black Hole Growth and ISM Conditions at Cosmic Noon

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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Mr. Gregory Robert Troiani (CoI)	University of Kansas Center for Research, Inc.
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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	FOUR_80_replan	NIRSpec MultiObject Spectroscopy	(2) MEGA_nirspec_catalog

ABSTRACT

We propose NIRSpec MSA observations of the EGS field in 4 pointings with the G140M, G235M, and G395M gratings covering a wavelength range of 0.9 - 5.0 microns. We will observe for 7965s in all gratings, designed specifically to find low luminosity AGN and probe the ISM of galaxies with SFRs $< 5 \text{ M}_{\text{sun}}/\text{yr}$ at $z=1-2$. All targets (130 primary + potentially 150 filler sources) have been selected from the MEGA survey, which covers the EGS field with MIRI imaging from 7.7 - 21.0 microns. The near-IR spectroscopy + mid-IR photometry will be combined to create a complete census of the ISM, star formation, and black hole growth within our targets. Our primary sample is selected from $z=0.5-5.0$ to: 1) confirm low luminosity or obscured AGN candidates (mid-IR selected) through high ionization lines such as [OIII]; 2) measure black hole masses (via H-beta) in unobscured AGN, down to $M(\text{BH}) = 10^7 \text{ M}(\text{sun})$; 3) measure metal content via lines such as [OII], [SII], [NII] and correlate with strength of PAH features; 4) use Halpha, Paschen-alpha to calibrate PAH SFR indicators in main sequence galaxies. Our filler sample is selected to: 1) confirm the AGN / obscured nature of little red dots; 2) distinguish between conflicting photometric redshift estimates. NIRSpec follow-up of MIRI-selected sources offers a unique window into the obscured assembly of mass within galaxies.

OBSERVING DESCRIPTION

We propose for 4 NIRSpec MOS pointings in the EGS field, covering the area that was included in the MEGA survey. We used the MSA planning tool to optimize the location of each pointing. We found that adding additional pointings does not significantly increase our number of primary targets, due to the significant size difference between the MIRI FOV and the NIRSpec FOV and the layout of the MEGA pointings.

In each pointing, we observing in the G140M/F100LP, G235M/F170LP, G395M/F290LP. All three gratings are necessary to enable both cosmic noon science (obtaining Hbeta, 0.468um, and Paschen-alpha, 1.875um) and high redshift observations (measuring Hbeta, [OIII], [NII], Halpha at $z \sim 5$). We use the recommended NRSIRS2 readout pattern and plan for 18 groups per integration, with 1 integrations per exposure, following the recommendation that groups are increased rather than integrations. The NRSIRS2 pattern is required due to the high data volume per visit. The ETC indicates that even our brightest targets will not saturate. As our target list is comprised of point sources and extended galaxies, we use a nod and dither strategy. We follow the recommended nod strategy of 3 positions in the slitlet. We ask for two dithers, with a dispersion of 5 shutters and cross-dispersion of 2 shutters. The dithers will also help estimate the amount of leakage between slits. 18 groups x 1 integration x 3 nods x 2 dithers gives a total exposure time of 7965.53 seconds per grating per pointing. For three gratings and four pointings, this gives a total exposure time of 26 hours.

We use the recommended 3 slitlet configuration. We allow for multiple targets per row, with a minimum separation of 80 slits. We found that this configuration would maximize the possible number of targets while minimizing leakage.

Target acquisition will utilize existing NIRCам images from the CEERS survey, which have the necessary pointing accuracy.

Proposal 7957 - Targets - MEGA Spectra: Black Hole Growth and ISM Conditions at Cosmic Noon

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	nirspec_catalog	RA: 14 19 37.4597 (214.9060821d) Dec: +52 52 38.46 (52.87735d) Equinox: J2000		
	Comments: Description=[]				
Fixed Targets	(2)	MEGA_nirspec_catalog	RA: 14 19 47.2154 (214.9467308d) Dec: +52 53 34.40 (52.89289d) Equinox: J2000		
	Comments: Description=[]				

Observation	Proposal 7957, Observation 1: FOUR_80_replan Tue Mar 31 21:01:34 GMT 2026 Diagnostic Status: Warning Observing Template: NIRSpec MultiObject Spectroscopy
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Proposal 7957 - Observation 1 - MEGA Spectra: Black Hole Growth and ISM Conditions at Cosmic Noon

Diagnostics

(FOUR_80_replan (Obs 1)) Warning (Form): Config c10 (#22) has 16 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c10 (#22) has 3 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c10 (#23) has 16 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c10 (#23) has 3 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c10 (#24) has 16 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c10 (#24) has 3 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c3 (#7) has 15 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c3 (#7) has 6 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c3 (#8) has 15 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c3 (#8) has 6 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c3 (#9) has 15 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c3 (#9) has 6 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c4 (#10) has 12 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c4 (#10) has 5 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c4 (#11) has 12 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c4 (#11) has 5 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c4 (#12) has 12 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c4 (#12) has 5 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c7 (#13) has 17 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c7 (#13) has 2 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c7 (#14) has 17 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c7 (#14) has 2 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c7 (#15) has 17 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c7 (#15) has 2 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c8 (#16) has 1 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c8 (#16) has 17 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c8 (#17) has 1 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c8 (#17) has 17 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c8 (#18) has 1 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c8 (#18) has 17 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c9 (#19) has 16 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c9 (#19) has 2 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c9 (#20) has 16 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c9 (#20) has 2 primary slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c9 (#21) has 16 filler slit traces affected by failed open shutters.
 (FOUR_80_replan (Obs 1)) Warning (Form): Config c9 (#21) has 2 primary slit traces affected by failed open shutters.
 (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
 (Visit 1:1) Warning (Form): The recommended value is 8 Reference Stars for this template.
 (Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
 (Visit 1:2) Warning (Form): The recommended value is 8 Reference Stars for this template.
 (Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
 (Visit 1:3) Warning (Form): The recommended value is 8 Reference Stars for this template.
 (Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Proposal 7957 - Observation 1 - MEGA Spectra: Black Hole Growth and ISM Conditions at Cosmic Noon

Fixed Targets	Name		Target Coordinates		Targ. Coord. Corrections				Miscellaneous		
	(2)	MEGA_nirspec_catalog	RA: 14 19 47.2154 (214.9467308d) Dec: +52 53 34.40 (52.89289d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: CLEAR; Readout: NRSRAPIDD2; 7 sources in 4 quads; [Reduced Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD2	3	1	4	343.577	
	2	Filter: F140X; Readout: NRSRAPID; 7 sources in 4 quads; [Reduced Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	3	Filter: F140X; Readout: NRSRAPIDD6; 7 sources in 4 quads; [Reduced Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
	4	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Reduced Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	Primary (482 sources)	Filler (133228 sources)	jwst-nirspec-mr	1.5			

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Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
	1	513314	214.862527	52.825834	23.3214	1	520279	214.785132	52.806928	23.2757
	1	516477	214.842987	52.828181	23.0912	1	522500	214.817950	52.841103	24.8512
	1	516935	214.794229	52.794530	22.3548	1	526959	214.760688	52.823138	24.4200
	1	520271	214.793828	52.810624	23.2881					
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
	2	502406	214.903177	52.793507	22.8189	2	512775	214.850052	52.812781	22.4303
	2	503659	214.883692	52.786784	21.5856	2	515779	214.813462	52.802987	22.1313
	2	505232	214.892176	52.800087	20.8519	2	516876	214.843476	52.829831	21.8063
	2	509788	214.814326	52.771997	22.0604					
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
	3	516011	214.901376	52.867211	22.9122	3	519053	214.885113	52.871227	22.6575
	3	518025	214.866477	52.852698	23.7617	3	528756	214.817988	52.882991	23.7560
	3	518347	214.869657	52.857125	23.3911	3	529971	214.844001	52.901816	22.3036
	3	518647	214.898665	52.878851	22.3904					
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
	4	516408	214.961879	52.910709	21.1750	4	528917	214.906511	52.946050	21.2321
	4	517911	214.991164	52.940336	22.1191	4	529737	214.913095	52.945518	21.8404
	4	526431	214.904315	52.922009	21.4190	4	530666	214.899331	52.934520	21.5838
	4	527328	214.910729	52.931463	21.6979	4	531351	214.913295	52.940741	21.2500

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Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	214.80942 Degrees 52.830408333333 34 Degrees	326.17528845963 16			3	3	3982.767
	2	2 (G235M/F170LP)	c1	3 Shutter Slitlet	214.80942 Degrees 52.830408333333 34 Degrees	326.17528845963 16			3	3	3982.767
	3	3 (G395M/F290LP)	c1	3 Shutter Slitlet	214.80942 Degrees 52.830408333333 34 Degrees	326.17528845963 16			3	3	3982.767
	4	1 (G140M/F100LP)	c2	3 Shutter Slitlet	214.80863833333 333 Degrees 52.830445833333 336 Degrees	326.17466660311 78			3	3	3982.767
	5	2 (G235M/F170LP)	c2	3 Shutter Slitlet	214.80863833333 333 Degrees 52.830445833333 336 Degrees	326.17466660311 78			3	3	3982.767
	6	3 (G395M/F290LP)	c2	3 Shutter Slitlet	214.80863833333 333 Degrees 52.830445833333 336 Degrees	326.17466660311 78			3	3	3982.767
	7	1 (G140M/F100LP)	c3	3 Shutter Slitlet	214.84367791666 665 Degrees 52.795813888888 89 Degrees	326.20256225069 426			3	3	3982.767
	8	2 (G235M/F170LP)	c3	3 Shutter Slitlet	214.84367791666 665 Degrees 52.795813888888 89 Degrees	326.20256225069 426			3	3	3982.767
	9	3 (G395M/F290LP)	c3	3 Shutter Slitlet	214.84367791666 665 Degrees 52.795813888888 89 Degrees	326.20256225069 426			3	3	3982.767
	10	1 (G140M/F100LP)	c4	3 Shutter Slitlet	214.84289687499 998 Degrees 52.795851666666 664 Degrees	326.20194116616 466			3	3	3982.767
	11	2 (G235M/F170LP)	c4	3 Shutter Slitlet	214.84289687499 998 Degrees 52.795851666666 664 Degrees	326.20194116616 466			3	3	3982.767
	12	3 (G395M/F290LP)	c4	3 Shutter Slitlet	214.84289687499 998 Degrees 52.795851666666 664 Degrees	326.20194116616 466			3	3	3982.767
	13	1 (G140M/F100LP)	c7	3 Shutter Slitlet	214.85820999999 999 Degrees 52.875861111111 114 Degrees	326.21407717181 523			3	3	3982.767
	14	2 (G235M/F170LP)	c7	3 Shutter Slitlet	214.85820999999 999 Degrees 52.875861111111 114 Degrees	326.21407717181 523			3	3	3982.767

Proposal 7957 - Observation 1 - MEGA Spectra: Black Hole Growth and ISM Conditions at Cosmic Noon

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
15	3 (G395M/F290LP)	c7	3 Shutter Slitlet	214.85820999999 999 Degrees 52.875861111111 114 Degrees	326.21407717181 523			3	3	3982.767
16	1 (G140M/F100LP)	c8	3 Shutter Slitlet	214.85742754166 665 Degrees 52.875898888888 89 Degrees	326.21345429944 387			3	3	3982.767
17	2 (G235M/F170LP)	c8	3 Shutter Slitlet	214.85742754166 665 Degrees 52.875898888888 89 Degrees	326.21345429944 387			3	3	3982.767
18	3 (G395M/F290LP)	c8	3 Shutter Slitlet	214.85742754166 665 Degrees 52.875898888888 89 Degrees	326.21345429944 387			3	3	3982.767
19	1 (G140M/F100LP)	c9	3 Shutter Slitlet	214.94671583333 334 Degrees 52.931625 Degrees	326.28454666986 244			3	3	3982.767
20	2 (G235M/F170LP)	c9	3 Shutter Slitlet	214.94671583333 334 Degrees 52.931625 Degrees	326.28454666986 244			3	3	3982.767
21	3 (G395M/F290LP)	c9	3 Shutter Slitlet	214.94671583333 334 Degrees 52.931625 Degrees	326.28454666986 244			3	3	3982.767
22	1 (G140M/F100LP)	c10	3 Shutter Slitlet	214.94593245833 335 Degrees 52.931663333333 33 Degrees	326.28392258661 677			3	3	3982.767
23	2 (G235M/F170LP)	c10	3 Shutter Slitlet	214.94593245833 335 Degrees 52.931663333333 33 Degrees	326.28392258661 677			3	3	3982.767
24	3 (G395M/F290LP)	c10	3 Shutter Slitlet	214.94593245833 335 Degrees 52.931663333333 33 Degrees	326.28392258661 677			3	3	3982.767
Special Requirements	Group Visits within 53.0 Days									
	Visits Same PA MSA Scheduled Aperture PA 326.2845 to 326.2845 Degrees (V3 187.70998 to 187.70998)									